

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

EC-7005(1)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Information Theory and Coding

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Make suitable assumptions wherever necessary.
1. a) Define information. State the various units of information and find relation between them. Show that entropy of a binary system (two events) is maximum when both events are equiprobable.
- b) A transmitter has an alphabet of four letters $[x_1, x_2, x_3, x_4]$ and the receiver has an alphabet of three letters $[y_1, y_2, y_3]$. The joint probability matrix is as follows:

	y_1	y_2	y_3
x_1	0.3	0.05	0
x_2	0	0.25	0
x_3	0	0.15	0.05
x_4	0	0.05	0.15

Calculate all entropies.

EC-7005(1)-CBGS

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[2]

2. a) Explain the source coding theorem to justify the fundamental limit on the average number of bits per source symbol.
- b) Define code efficiency. Determine the coding efficiency by applying the Huffman coding method for the following message resemble:
- $$[X] = [x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7 \quad x_8]$$
- $$[P] = [1/4 \quad 1/8 \quad 1/16 \quad 1/16 \quad 1/16 \quad 1/4 \quad 1/16 \quad 1/8]$$

3. a) Design a block code with a minimum distance of 3 and a message block size of 8 bits.
- b) Consider a (7, 4) block code generated by:

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & : & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & : & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & : & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & : & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{matrix} & & & & & I_4 & & P \end{matrix}$$

Explain the errors syndrome S helps in correcting a single error.

4. a) Explain the Hamming distance and its applications.
- b) Explain how the error is detected at decoder circuit in case of linear block codes.
5. a) The generator polynomial of a (7, 4) cyclic code is $g(x) = 1 + x + x^2$. Find the code-words of this code.
- b) Given that Message (D) = 1010001101 and Pattern (P) = 110101, calculate the FCS (R) and the message which is actually Transmitted (T) using Cyclic Redundancy Check method. Also check it for correctness at receiver end that received message is correct or not. Do all calculations in clear manner.

EC-7005(1)-CBGS

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[3]

6.
 - a) What are Convolution codes? Explain encoding and decoding for convolution codes.
 - b) Explain Viterbi algorithm for maximum likelihood decoding.
7. A convolution encoder is defined by the following generator polynomials:
$$g_0(x) = 1 + x + x^2 + x^3 + x^4$$
$$g_1(x) = 1 + x + x^3 + x^4$$
$$g_2(x) = 1 + x^2 + x^4$$
 - a) What is the constraint length of this code?
 - b) How many states are in the trellis diagram of this code?
 - c) What is the code rate of this code?
8. Write short notes on any two of the following:
 - i) Trellis Code
 - ii) Binary Erasure Channel
 - iii) Entropy
